Solving Systems of Equations

Recall: Two Methods:

1. SUBSTITUTION
2. ELIMINATION

Example 1: Solve the system.

$3x + 5y = 11$ ①
$2x - 3y = 1$ ②

using elimination:

① × 2 $6x + 10y = 22$ ③
⊖ ② × 3 $6x - 9y = 3$ ④

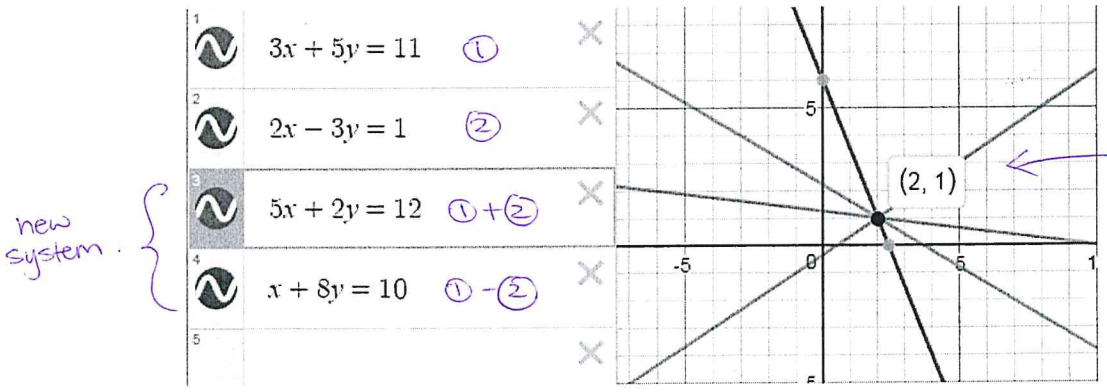
"New Equivalent System"

$19y = 19$
$y = 1$

Sub $y = 1$ into ①
$3x + 5(1) = 11$
$3x = 6$
$x = 2$

∴ P.O.I (2, 1)

Note: To solve a system by elimination, we replace the original system with a new EQUIVALENT system that has the same solution.



new system.

Same solution/P.O.I for all systems.

To create equivalent system, you may replace any equation with:

1. a scalar multiple of itself

e.g. take ① × 2 multiply every term by scalar of 2.
→ give same/coincident line or plane.

2. a linear combination of itself and any other equation

e.g. example of picture above.
① + ② add 2 equations to get a new equation.
→ a different line/plane but

What does the P.O.I / solution of this system mean?

Example 2: Solve the following system.

$$x + 2y - 4z = -7 \quad ①$$
$$2x + 5y - z = 9 \quad ②$$
$$3x - y + 2z = 7 \quad ③$$

Pick a variable to eliminate. e.g. x!

Using ① & ②

$2 \times$ ① $\quad 2x + 4y - 8z = -14$

$-$ ② $\quad 2x + 5y - z = 9$

$-y - 7z = -23$.

OR $y + 7z = 23$ ④

Using ① & ③

$3 \times$ ① $\quad 3x + 6y - 12z = -21$

$-$ ③ $\quad 3x - y + 2z = 7$

$7y - 14z = -28$

OR $y - 2z = -4$ ⑤

Equivalent System

eliminated the x.

$$x + 2y - 4z = -7 \quad ①$$
$$0x + y + 7z = 23 \quad ④$$
$$0x + y - 2z = -4 \quad ⑤$$

} subtract to get another new system.

Now use ④ & ⑤ to eliminate y!

Equivalent System:

$$x + 2y - 4z = -7 \quad ①$$
$$0x + y + 7z = 23 \quad ④$$
$$④ - ⑤ \quad 0x + 0y + 9z = 27 \quad ⑥$$

From ⑥ $9z = 27$, $z = 3$

Sub into ④ $y + 7(3) = 23$, $y = 2$

Sub into ① $x + 2(2) - 4(3) = -7$, $x = 1$

∴ P.O.I. $(1, 2, 3)$

Example 3: Solve the following system.

$$3x + 5y - z = 6 \quad ①$$
$$2x + y + 4z = 4 \quad ②$$
$$6x + 10y - 2z = 12 \quad ③$$

Using ① & ②

$2 \times$ ① $\quad 6x + 10y - 2z = 12$

$-$ $3 \times$ ② $\quad 6x + 3y + 12z = 12$

$7y - 14z = 0$. ④

Using ① & ③

$2 \times$ ① $\quad 6x + 10y - 2z = 12$

$-$ ③ $\quad 6x + 10y - 2z = 12$

$0z = 0$.

} same plane!!

Equivalent system

$$3x + 5y - z = 6 \quad ①$$
$$0x + 7y - 14z = 0 \quad ④$$
$$0z = 0 \quad ③$$

$z \in \mathbb{R}$!! Let $z = t$

Sub into ④ $7y - 14t = 0$, $7y = 14t$, $y = 2t$

Sub into ① $3x + 5(2t) - (t) = 6$, $3x + 9t = 6$, $3x = 6 - 9t$, $x = 2 - 3t$

∴ $(x, y, z) = (2 - 3t, 2t, t)$

$= (2, 0, 0) + t(-3, 2, 1)$, $t \in \mathbb{R}$

The intersection is a LINE!!